Phase-Wave Differential Calculus: Neural Tuning, Variability Tokens, and Taylor-Sequence Approximation in Oscillatory Neural Networks

Author: Micah Blumberg

Self Aware Networks Research Institute

https://selfawarenetworks.com

July 2026

Abstract

The same neural event can matter differently depending on what a receiving cell or population is already doing. Existing accounts measure important parts of that problem - rate, phase, joint phase-rate, prediction error, phase response, plasticity, waveform shape, or nonlinear history - but they do not by themselves specify one typed record that joins the departure, its receiver, its physical route, and the receiver's subsequent change. This paper defines a **phase-wave differential** (PWD) as that bounded, receiver-relative record. Its candidate components are phase, event rate, amplitude, duration, transmitted quantity, variability, and declared spatial or network context, each measured against a frozen tonic reference. A measured difference qualifies as a candidate biological token only if a named receiver preserves or transforms it, a selective perturbation changes the predicted consequence, and the representation adds held-out value beyond matched alternatives.

The record becomes numerical only through a frozen measurement contract that declares domains, estimators, units, normalization, missingness, feature signature, target, nuisance controls, comparator, and failure rule before outcome inspection. Coefficient of variation is restricted to suitable positive variables; circular statistics handle phase; covariance handles multivariate structure. Neural Tuning is written as a receiver-state update. Taylor jets provide a local approximation language, while Volterra and recurrent state-space models remain mandatory comparators when history matters. A bounded electrical component is also specified: signed source activity passes through an explicit complex transfer to a named receiver, is projected onto the receiver's reference, and is differenced from a cross-fitted tonic expectation. This prevents amplitude, synchrony, geometry, delay, inhibition, and receiver sensitivity from being collapsed into one opaque "coherence" score.

The genealogy is staged: a public April 2017 Neural Lace Podcast posed a multimode neural-communication problem and named Alternating Transmission Protocol; a public July 2022 Git note related successive phase intervals to changing vectors and Taylor-series-like approximation; public 2022 videos articulated a receive-transform-project array chain and multidimensional event features; and a public September 2024 mixed dialogue joined tonic departure, transmitted magnitude, receiver consequence, derivative-like change, and Neural Tuning. These are provenance anchors, not biological validation.

Five deterministic synthetic analyses test the implementation. Three designed fixtures verify typed-variable recovery, grouped receiver dependence, and a persistent post-washout target. Two harder comparisons are adverse: PWD-conditioned ARX wins `0/9` in-distribution and `0/9` shifted regimes against Volterra and ordinary ARX, while typed PWD misses both frozen NAPOT reconstruction thresholds and the raw recurrent model performs best. The surviving contribution is therefore a reproducible measurement, comparison, and falsification program - not generic approximation superiority, biological validation, or an established NAPOT reconstruction advantage.

Keywords: phase-wave differential; neural tuning; coefficient of variation; circular variance; Taylor approximation; Volterra kernels; receiver-relative coding; oscillatory neural networks; causal signal processing

Scope and evidence boundary

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1. Research question and paper-level claim

Imagine that a visual or bodily change arrives while a neural population is already active. The event does not enter an empty receiver. Membrane state, inhibition, recent activity, oscillatory timing, anatomical route, and learned sensitivity affect what the same source event can do. The scientific problem is therefore not only to record that "a spike happened" or that power in a frequency band changed. It is to state what changed relative to a named receiver's ongoing condition, what physical quantity traveled along which route, and whether the receiver subsequently changed its output, tuning, routing, reconstruction, or behavior.

Established neuroscience supplies several necessary descriptions. Rate coding measures how often events occur. Phase coding measures timing relative to an oscillation. Joint phase-rate models combine those variables. Predictive-processing models ask whether an event departs from an expectation. Phase-response curves measure how a perturbation changes an oscillator. Spike-timing-dependent plasticity and other plasticity rules test persistent effects. Volterra and state-space models represent nonlinear history. Each is a required comparator; none is automatically interchangeable with the complete receiver-relative operation considered here.

In plain language, the proposed analysis performs seven steps:

```
name the receiver and its expected ongoing state
-> bound the new event in time
-> measure each physical departure in its proper domain
-> record the route and the receiver's state
-> test whether the receiver preserves or transforms the distinction
-> test whether later tuning or routing changes
-> compare the full account with simpler and history-aware alternatives
```

This bounded receiver-relative record is called a **phase-wave differential**. The term is earned by the operation above: it is not a new name for phase coding, traveling waves, prediction error, synchrony, or coefficient of variation. Phase is one field in the record; a traveling wave is one possible spatiotemporal carrier; prediction error is one possible interpretation; synchrony is one possible relation; and variability requires a domain-appropriate statistic. The paper-level question is whether the complete, frozen record predicts a named consequence beyond those component accounts without expanding after a negative result.

2. Relationship to the operational companion empirical estimator

This manuscript and the separate paper *Phase-Wave Differentials: A Receiver-Relative Signal Operator for Testing Neural State Updates* answer different questions. the companion empirical paper defines a minimal four-component LFP estimator: wrapped phase departure, log-amplitude departure, causal duration, and a coordinate-normalized lagged phase gradient. It freezes one Allen Neuropixels benchmark for that estimator.

The present paper asks whether a larger typed calculus can connect event variability, transmitted quantity, receiver state, tuning updates, and successive approximation. The the companion empirical paper vector is therefore one candidate measurement subtype, not the full definition of PWD. the companion empirical paper Draft 5 now reports the frozen development procedure across 14 coordinate-eligible Allen Neuropixels sessions, 921 eligible VISp units, three fixed projection seeds, and 908,295 causal samples. Every session-level primary increment was negative: the cross-session median raw-minus-PWD held-out Poisson-deviance difference was `-0.001099`, with a descriptive session-bootstrap 95% interval from `-0.001301` to `-0.000872` and an exact two-sided sign test of `0/14` positive sessions (`p=0.000122`). The five sealed final-test sessions remain unopened. This is an adverse biological-development result for the tested tested 25-feature implementation; it is neither a test of the broader broader calculus record nor evidence for it. Broad language in this paper cannot rescue the companion empirical estimator by changing components after outcomes are inspected, and this paper cannot treat its designed synthetic fixtures as independent confirmation of the companion empirical estimator. Every tested PWD version requires a frozen signature, baseline, receiver, target, comparator, and failure rule.

This separation prevents two opposite errors: collapsing the historical SAN construct into one scalar circular residual, and expanding PWD after the fact until no result can falsify it.

3. Historical donor and source boundary

The principal 2024 historical donor is:

- `02san.md`

The September 8, 2024 file contains the following relevant authored proposal family:

- a PWD is an unexpected spike, burst, inhibition, or transmitted-quantity departure from a tonic group relation;
- duration and amplitude contribute to transmitted magnitude;
- the event changes synaptic frequency or later functional relations;
- the difference is compared with a derivative or loss-like term;
- events are discrete even when repeated oscillation looks continuous at a coarser scale; and
- Neural Tuning links changed cellular activity to changed functional connectivity.

The file is a mixed working dialogue containing both Micah-authored assertions and generated explanatory material. It establishes concept genealogy, not biological validation. This manuscript attributes the historical route only to passages explicitly headed "Micah Blumberg said"; generated elaboration is not presented as an independently dated Micah claim. Exact message-level locators, commit identity, hash, date type, and source grade are recorded in the paper-specific Claim-Source-and-Genealogy Atlas.

3.1 The 2017 protocol question and Alternating Transmission Protocol

The April 2017 *Neural Lace Podcast* asked whether communication among neurons, brain regions, and oscillatory activity is better compared with connection-and-feedback transmission or connectionless packet transmission [21]. The computer-network comparison used TCP and UDP as functional prompts, not as claims that the brain literally implements Internet protocols. The proposed **Alternating Transmission Protocol** (ATP) could switch between feedback-dependent and connectionless modes. Its acronym was deliberately chosen to echo adenosine triphosphate, the cellular energy currency, joining a communication question to the energetic cost of biological signaling.

This 2017 source establishes an earlier problem statement: identify a multimode brain-network protocol that could eventually support read / write neural interfaces. It does not contain the later PWD measurement contract, seven-component typed record, Taylor / Volterra comparison, or NAPOT reconstruction model. The chronology is therefore:

```
2017 protocol question and ATP hybrid
-> 2022 phase-specific and Taylor-sequence proposals
```

```
-> 2024 typed PWD and Neural Tuning proposal
-> present frozen measurement and falsification program
```

The sequence preserves continuity without backdating later mathematics or treating an acronym analogy as biological evidence.

3.2 Staged Taylor and typed-PWD genealogy

The earliest explicit Taylor route currently verified in the SAN Git history is `a0258z`, created in commit 560df7f84aa66c232c89e3074bd4a2a287334f26 on July 5, 2022 and renamed to `a0258z.md` in commit 71bb b1aa7cb672d15fbc6e3abf3487fefe4dc8b8 later that day. The originating text describes each action-potential phase interval as a changing vector, successive events as Taylor-series-like polynomial terms, inhibition as modification of later terms, and the resulting sequence as a progressively refined curve or represented pattern. The note also connects the mechanism to sensory representation and coordinated movement. This is a historical approximation proposal, not a proof that a neural circuit computes a Taylor expansion.

The later donor `02san.md` entered the Git record in commit 64f018d2d93b7de6718058c4724c43b40e5d d4b7 on September 6, 2024 and was updated in commit df4d7f8d3270d56de167035c4987c3e072696c79 the same day. Its current local SHA-256 is ADD0F593BC0BA289907749D2D290A59556410046D7 203A2F94DFEB2CDC6D8706. Micah-authored passages define PWD as an unexpected spike, burst, inhibition, or transmitted-quantity departure from a tonic group relation; connect amplitude and duration to transmitted magnitude; compare the differential with a physical derivative or loss-like term; insist that apparently continuous oscillation is composed of bounded events; and connect those events to Neural Tuning and changed routes.

`02san.md` is a mixed dialogue. Headings such as "Micah Blumberg said" and "Self Aware Networks GPT said" are substantive authorship boundaries. Generated responses can explain later manuscript development but cannot be cited as independently dated Micah claims or as scientific evidence. The present paper uses the authored passages as genealogy and independent literature as the evidentiary layer.

3.3 Electrical-valence and action lineage

The receiver-projected electrical component also has a staged history. The Git-fixed `a0271z.md` note of August 10, 2022 explicitly connected changes in phase signals, "relative electrical valence," neural-array transmission, and behavior. That source establishes the phrase and the proposed conceptual join. It does not establish that psychological valence equals voltage, that a local field potential measures net synaptic current, or that the 2026 transfer model already existed.

The Git-fixed `a0111z.md` note of August 19, 2022 excerpts the bat fronto-auditory study's pre-vocal directional-flow findings and preserves the authors' statement that the relationship remained correlational. The September 14, 2022 `a0001z.md` note then interprets the bat route as an example of top-down control with PWDs. These records establish Micah's 2022 problem framing: coordinated oscillatory departures may accumulate into route-specific action. They do not demonstrate causal bat vocal control or identify the present equations.

Equations 4-6 are the 2026 mathematical operationalization of that earlier conceptual route. Their explicit sign, complex transfer, receiver projection, and cross-fitted tonic reference were added after an audit found that the simpler scalar $\sum_i w_{ic} I_i \cos(\phi_i - \Phi_c)$ hid anatomy, delay, geometry, sensitivity, and sign inside one weight. The later correction is additive and intent-preserving: the 2022 phase-to-behavior genealogy remains visible, while the stronger identifiable model is dated to its actual formalization.

4. Verified 2022 source routes

The historical program is not inferred only from the 2024 donor. Four public 2022 video transcripts have been deterministically segmented and atomized. The locators below establish Micah's proposal genealogy; they are not biological validation.

Route	Exact 2022 locators	Source-faithful proposition	Scientific boundary
Array-to-array transformed representations	July 28 `S0015–S0017`; July 31 `YT31-A08`, `S0116–S0143`, `S0202–S0207`, `S0258–S0260`, `S0457–S0459`	An upstream array collects a pattern, a neuron integrates it, fan-out projects a signal pattern, and the downstream array receives and transforms it.	Strong authored architecture anchor. Fan-out is background biology; "screen" and "picture" remain functional analogies until operationalized.
Cortical-column assumptions	July 31 `YT31-A29`, `S0202–S0206`; September 18 `YT5-A17`, `S0211–S0224`	Cortical columns were provisionally treated as oscillatory stages whose cooperation can be selected by synchrony and inhibition.	The July transcript explicitly labels the column model an assumption. The manuscript does not claim one universal cortical-column algorithm or attribute that claim to Numenta.
Object and location coding	July 31 `YT31-A13`, `S0211–S0226`, `S0249–S0263`, `S0286–S0289`; September 18 `YT5-A26`, `S0418–S0424`	Object, feature, body, and location relations motivate a multidimensional receiver state and volumetric scene proposal.	These segments establish SAN genealogy, not a literal 3D neural grid or proof that one event contains a complete object.
Apical / basal evidence-context comparison	September 18 `YT5-A24`, `S0357–S0365`, `S0404–S0421`; `YT5-A27`, `S0424–S0439`	Basal forward input and apical / back-propagating context were proposed to interact in burst selection and recurrent rendering.	This is a bounded SAN hypothesis. Basal does not universally mean evidence, apical does not universally mean expectation, and the joined burst mechanism requires direct testing.
Distributed rendering without a hidden viewer	July 31 `YT31-A15`, `S0262–S0268`, `S0296–S0307`, `S0430–S0438`; September 18 `YT5-A05`, `S0050–S0056`, `S0103–S0109`	The same distributed cells and assemblies that generate recurrent patterns also detect the looped outputs; no localized inner spectator receives a finished picture.	Strong anti-homuncular genealogy anchor. It does not establish phenomenological sufficiency or a consciousness criterion.

The custody ledgers are july28-july30-full-atom-ledger-20260719.md, july31-full-atom-ledger-20260719.md, and september18-full-atom-ledger-20260719.md under the governed SAN 2022 segment index. The July 31 and September ledgers preserve transcript-corruption and anatomical-correction warnings. Those warnings govern interpretation: rough or auto-transcribed language is not silently upgraded into polished scientific fact.

5. Operational definitions

Tonic reference state

For receiver `r`, define a reference state over a declared estimation window `W_R`:

$$\mathbf{b}_r(t; W_R) = \left[\bar{\phi}_r, \bar{f}_r, \bar{A}_r, \bar{\tau}_r, \bar{q}_r, \bar{\mathbf{x}}_r, \mathbf{\Sigma}_r\right]. \tag{1}$$

This is not a universal brain rhythm. It is the expected or maintained state of a named receiver under a specified preparation, task, behavioral state, reference signal, and timescale.

Bounded PWD event

Event `i` begins at `t_i^-` and ends at `t_i^+`. Its candidate feature record is:

$$\mathbf{p}_{r,i} = [\Delta\phi_{r,i}, \Delta f_{r,i}, \Delta A_{r,i}, \Delta\tau_{r,i}, \Delta q_{r,i}, \Delta V_{r,i}, \Delta\mathbf{x}_{r,i}] \,. \tag{2}$$

The variables remain typed:

- `Delta phi`: circular timing difference relative to a declared reference;
- `Delta f`: change in a declared oscillation or event-rate estimate;
- `Delta A`: change in a measured amplitude variable;
- `Delta tau`: event-duration change;
- `Delta q`: change in transmitted quantity, measured at a specified synapse or population rather than inferred automatically from a field trace;
- `Delta V`: change in an appropriate variability statistic; and
- `Delta x`: spatial, anatomical, population-state, or network-position variables declared before analysis.

Frequency is related to phase evolution and should not be treated as an independent degree of freedom unless the estimator and window justify it. Amplitude, extracellular voltage, transmitter amount, and postsynaptic effect are different biological variables.

Measurement contract and numerical encoding

Equation 2 is a typed record, not automatically one Euclidean vector. Circular phase, positive intervals, event rates, voltages, physical transmission quantities, variability statistics, anatomical labels, and graph or population coordinates do not share a metric merely because they appear in one list. Before fitting a numerical model, analysis version `v` must freeze a measurement contract

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M_v = {receiver, reference signal, event and reference windows, field domains,
       estimators, units, normalization, missingness, feature signature,
       target, grouping, nuisance controls, comparator, and failure rule}.
```

The contract supplies an encoding

$$\mathbf{z}^{(v)}_{r,i} = E_{\mathcal{M}_v}(\mathbf{p}_{r,i}) \in \mathbb{R}^{d_v}, \tag{3}$$

which may use sine / cosine phase coordinates, standardized positive variables, declared categorical or graph encodings, interaction terms, and explicit missingness indicators. A norm, distance, covariance, or scalar collapse applies to `z`, not to the unencoded heterogeneous record. The context field `Delta x` is therefore closed under the frozen signature: an unscheduled anatomical, behavioral, or network variable cannot be added after a failed outcome and still count as the same test.

`Delta q` likewise requires a route-specific observation model. It may denote directly measured or independently calibrated transmitter release, charge transfer, or another declared physical quantity. Field amplitude and amplitude-times-duration are not automatically transmitted quantity. When `q` is not observed, the field remains missing rather than being silently inferred from a convenient waveform proxy.

In this paper, *calculus* names the candidate sequence of reference construction, typed differencing, encoding, receiver-conditioned updating, model comparison, and perturbation-rescue testing. No theorem here establishes a closed algebra, unique encoding, convergence of the update operator, or generic predictive superiority.

Receiver-projected electrical component

A further problem arises when several source populations reach one candidate receiver at nearly the same time. Their contributions cannot be ranked from phase alignment or spectral power alone. Excitatory and inhibitory sign, amplitude, propagation delay, geometry, orientation, coupling, and receiver sensitivity can make apparently aligned sources reinforce, cancel, or affect a different route. Communication-through-coherence supplies an established component hypothesis and comparator [22], but the present question is more specific: what signed contribution reaches this receiver, and how unexpected is it relative to this receiver's ongoing condition?

For candidate receiver or assembly `c`, let source `i` have nonnegative analytic amplitude $A_i(t)$, phase $\phi_i(t)$, and explicit excitatory or inhibitory sign $\sigma_{ic} \in \{-1, +1\}$. Let $G_{ic}(t) = g_{ic}(t)e^{-\mathrm{i}\psi_{ic}(t)}$ be a declared complex transfer factor whose magnitude includes calibrated coupling and receiver sensitivity and whose angle includes propagation delay, orientation, measurement reference, and other calibrated phase shifts. The effective complex source sum is

$$Z_c(t) = \sum_{i=1}^{N_c} \sigma_{ic} G_{ic}(t) A_i(t) e^{\mathrm{i}\phi_i(t)}. \tag{4}$$

Every summand must be calibrated into the same declared receiver-domain units before the sum is formed. If sources are available only in incomparable modalities or arbitrary sensor units, they remain separate typed fields rather than being added.

If $\Phi_c(t)$ is the named receiver's directional or phase reference, the signed receiver projection is

$$D_c(t) = \mathrm{Re}\left[e^{-\mathrm{i}\Phi_c(t)} Z_c(t)\right]. \tag{5}$$

The tonic reference must be conditional and learned without held-out outcome leakage. For preregistered context X_t, define the training-fold expectation and held-out electrical departure as

$$\begin{aligned} m_c(X_t) &= \mathbb{E}[D_c(t) \mid X_t], \\ \mathrm{PWD}_{e,c}(t) &= D_c(t) - \widehat{m}_c^{(-k)}(X_t), \end{aligned} \tag{6}$$

where $(-k)$ identifies a model fitted without the held-out group. This operation earns the SAN label **receiver-relative electrical valence**: a signed, coherence-sensitive effective drive relative to a tonic expectation. It is one candidate field in the typed PWD record. It is not bulk net-charge accumulation, chemical valence, muscular energy, psychological valence, or a complete choice variable. If G_{ic} is not measured or independently constrained, D_c is a latent model quantity and cannot be advertised as an identified current.

The decomposition also prevents "more coherence" from being treated as automatically "more drive." Coherent sources can cancel when their effective signs or orientations oppose, while a structured splay state can exert a reliable receiver-specific influence. Any biological test must therefore compare the full projection with rate-only, power-only, phase-only, geometry-free, and capacity-matched recurrent models;

control movement, arousal, body state, task history, filtering, and common input; and estimate every nuisance transformation inside training folds.

Equations 4-6 are a newly specified successor interface. They do not reinterpret the adverse companion-paper development result or the adverse this paper approximation and reconstruction fixtures. The complete transfer-identifiability, motor, bat-vocalization, loss, rescue, and returned-sensory-consequence program belongs to the companion electrical-valence / motor-choice paper; broader calculus records only the typed-record interface and its measurement firewall.

PWD code criterion

A feature vector is not yet a token merely because it can be measured. It becomes a candidate biological token only when a named receiver preserves or transforms the distinction, selective perturbation changes the predicted consequence, and the representation adds held-out value beyond matched alternatives.

6. Coefficient of variation and variability statistics

For positive inter-event intervals:

$$CV_{ISI} = \frac{\sigma_{\Delta t}}{\mu_{\Delta t}}. \tag{7}$$

Ordinary CV is unstable when the mean approaches zero, is inappropriate for signed variables with arbitrary zero points, and is not a circular phase statistic. The analysis therefore compares:

- global interspike-interval `CV`;
- adjacent-interval `CV2`;
- local variation `LV`;
- Fano factor for count variability;
- circular variance or resultant length for phase;
- trial-to-trial covariance of the full PWD vector; and
- conditional variability after controlling stimulus, movement, arousal, and history.

Shinomoto, Shima, and Tanji showed that local interval variation can distinguish cortical-neuron firing-pattern classes in awake macaque recordings (2003). This supports local variability as a measurable neural property. It does not establish PWD or semantic tokens.

7. Neural Tuning as the update operator

Let the receiver's input-output transformation be:

$$\mathbf{y}_{r,n} = F_r(\mathbf{u}_n, \mathbf{c}_n; \boldsymbol{\theta}_{r,n}), \tag{8}$$

where `u` is input, `c` is current context, and `theta` is the receiver's learned and current state. A PWD-conditioned tuning update is:

$$\boldsymbol{\theta}_{r,n+1} = U_r(\boldsymbol{\theta}_{r,n}, \mathbf{p}_{r,i}, \mathbf{c}_n, \mathbf{y}_{r,n}). \tag{9}$$

The analysis distinguishes three outcomes:

1. immediate state dependence without learning;
2. changed effective connectivity or routing during a task; and
3. persistent plastic change.

Functional connectivity is an analysis-dependent statistical or effective relation, not a physical synapse created whenever a frequency changes. A tuning claim must name the route, receiver, timescale, intervention, and persistence criterion.

8. Taylor sequence and Volterra boundary

For a smooth finite-dimensional receiver model near operating point `z_0`, successive Taylor polynomials are:

$$P_K(\Delta\mathbf{z}) = \sum_{k=0}^{K} \frac{1}{k!} D^k F_r(\mathbf{z}_0)[\Delta\mathbf{z}^{\otimes k}]. \tag{10}$$

The sequence P_0,\P_1,\...,\P_K is the proposed **Taylor Sequence of Polynomials**. A temporally ordered PWD stream may provide samples from which a receiver or external model estimates progressively higher-order local relations. The biological system is not assumed to write symbols, factorials, or polynomial coefficients.

Because neural responses depend on event history, a simple Taylor expansion may be insufficient. A Volterra functional expansion is the more direct comparator for nonlinear systems with memory:

$$y(t) = h_0 + \sum_{k=1}^{K} \int \cdots \int h_k(\tau_1, \ldots, \tau_k) \prod_{j=1}^{k} u(t - \tau_j) d\boldsymbol{\tau} + \epsilon(t). \tag{11}$$

Volterra methods have already been used to identify nonlinear biological and neuronal input-output transformations. Korenberg and Hunter review the conditions and limitations of such approximations (1996); Zanos and colleagues applied multi-input nonlinear models to neuronal ensembles (2008); and point-process Volterra kernels have been applied to hippocampal spike transformations (2015).

The candidate mathematical novelty is the typed PWD event representation and its receiver-relative update semantics, not the existence of polynomial or Volterra expansions.

9. Distinction from neighboring theories

Framework	Core measured relation	What PWD adds as a hypothesis
Rate coding	Count or rate varies with condition	Typed departure from an ongoing receiver state
Phase coding	Event timing relative to a cycle carries information	Phase is one component inside a route- and receiver-specific event
Phase-rate coding	Joint phase and rate improve decoding	Adds duration, magnitude, variability, spatial state, receiver use, and tuning consequence

Framework	Core measured relation	What PWD adds as a hypothesis
Communication through coherence	Relative timing can make interpopulation communication more or less effective	Requires explicit sign, transfer, receiver projection, conditional tonic reference, and held-out incremental consequence rather than treating coherence as drive by itself
Predictive coding	Prediction-error relations update a model	Not every PWD is a prediction error; PWD also covers unexpected inhibition, routing, waveform, and transmitted-quantity changes
Phase-response curve	Perturbation phase predicts oscillator timing shift	PWD asks whether the resulting multidimensional event changes a downstream receiver and tuning relation
STDP	Relative spike timing changes synaptic efficacy under bounded rules	One candidate persistent update mechanism, not the definition of PWD
Volterra / Wiener model	Nonlinear history-dependent input-output approximation	A strong mathematical baseline and possible estimator for PWD-conditioned transformations

Montemurro and colleagues demonstrated phase-of-firing information beyond spike count in macaque V1 (2008). That is a required baseline, not proof of the broader PWD operator.

10. Atomized comparison appendix

The comparison is made at the component level. A prior source can support a phase, timing, prediction, plasticity, traveling-wave, or nonlinear-identification atom without supplying the joined joined operator.

Framework and source	Inherited atom	What the source establishes	What it does not establish	Additional term and discriminator
Phase-of-firing coding: Montemurro et al. [3]; Huxter et al. [14]; Voloh et al. [15]	Spike timing relative to an ongoing rhythm can add information beyond rate in bounded preparations.	Phase and rate can be separable or jointly informative; phase-specific firing can encode task variables.	A seven-component PWD, transmitted quantity, receiver-specific tuning, or semantic token status.	Freeze the typed vector and require incremental held-out value beyond rate and phase-rate baselines.
Communication through coherence: Fries [22]	Temporally coordinated source activity can alter effective communication with postsynaptic targets.	Synchronization is a plausible and testable component of selective neuronal communication.	That coherence equals signed receiver drive, that aligned sources cannot cancel, or that a PWD has been measured.	Estimate explicit sign and transfer, project onto a named receiver, subtract a cross-fitted tonic expectation, and ablate amplitude, alignment, geometry, and recurrent-state alternatives.
Local firing variability: Shinomoto et al. [4]	Local interval statistics can distinguish firing-pattern classes.	Variability is measurable and ordinary global CV is not the only statistic.	That CV defines PWD or that one variability measure applies to phase, signed voltage, and intervals.	Use domain-appropriate interval, circular, count, and covariance statistics; ablate each before interpretation.
Predictive coding: Rao and Ballard [16]	Feedback predictions and feedforward residual errors can be modeled in a hierarchy.	Prediction error is a strong comparator for unexpected sensory activity.	That every PWD is a prediction error or that prediction error necessarily changes a named receiver's persistent tuning.	Compare typed events directly with capacity-matched prediction-error features and require receiver-specific consequences.

Framework and source	Inherited atom	What the source establishes	What it does not establish	Additional term and discriminator
Phase-response curves: Gutkin, Ermentrout, and Reyes [17]	Perturbation phase can predict the timing shift of an oscillator.	A phase-dependent response is experimentally measurable for a declared oscillator and perturbation.	Multidimensional transmitted magnitude, route selection, or persistent plasticity.	Require PWD components beyond phase to improve held-out response and selective perturbation-rescue tests.
Spike-timing-dependent plasticity: Bi and Poo [18]	Relative pre / post timing can determine persistent potentiation or depression under bounded conditions.	One concrete timing-sensitive persistent-update mechanism.	A universal learning rule, an immediate routing effect, or the definition of PWD.	Separate immediate response, effective routing, and persistent tuning, with an explicit washout and persistence criterion.
Traveling cortical waves: Muller et al. [19]; Aggarwal et al. [20]	Stimulus-evoked activity can propagate across cortical space; feedforward and feedback waves can differ in frequency and direction.	Spatiotemporal wave structure and phase relations are measurable in specific preparations.	A global orthogonal lattice, a conscious display, or one universal tonic / phasic band assignment.	Treat spatial context as a declared measurement and test whether it adds receiver-specific prediction beyond local activity and common-input controls.
Nonlinear system identification: Korenberg and Hunter [5]; Zanos et al. [6]; Sandler et al. [7]	Volterra and related dynamic models approximate nonlinear systems with history.	Strong history-aware baselines exist and can outperform static local expansions.	PWD semantics or a biological token.	Simulation 4 makes these mandatory comparators and retires generic PWD approximation superiority after the adverse result.
Phase synchrony and waveform controls: Lachaux et al. [8]; Cole and Voytek [9]	Phase relations require explicit estimators; waveform shape can generate misleading spectral structure.	Estimator, reference, filtering, and waveform choices are load-bearing.	Receiver meaning, causal routing, or consciousness.	Freeze estimator and nuisance controls, then demand held-out receiver consequence and selective loss / rescue.

The joined candidate contribution is therefore narrow: a receiver-relative typed event, a declared reference and receiver, an update target, and a falsification rule. This paper does not claim ownership of phase, temporal coding, prediction error, plasticity, traveling waves, Taylor series, or Volterra identification.

11. Simulation program and current results

Simulation 1: Identifiability

Generate coupled conductance-based or validated reduced oscillator networks with known perturbations. Test whether the fitted PWD vector recovers the manipulated variables without conflating phase, frequency, amplitude, duration, and filtering artifacts.

Draft 2 fixture and result. A deterministic linear-nonlinear generator produced 6,000 events with independently manipulated circular phase, frequency departure, amplitude departure, duration departure, and transmitted-quantity proxy. The observation model added phase-frequency cross-talk, amplitude-duration filtering, quantity cross-talk, and noise. A common reference was subtracted with wrapped circular arithmetic. The split contained 4,200 training and 1,800 held-out events.

The seven-feature typed model used sine and cosine of relative phase, observed frequency, amplitude, duration, quantity, and the declared filter nuisance. It achieved per-target held-out R-squared values 0.9832,\0.9844,\0.9868,\0.9848,\0.9852,\0.9848 in the order sine-phase, cosine-phase, frequency, amplitude, duration, and quantity, for a mean of 0.984844. The three-feature phase-rate baseline achieved 0.492430 mean R-squared because it recovered the variables it contained but not the omitted magnitude and duration variables. A collapsed norm achieved 0.004528. Shuffled training labels produced -0.005857 mean R-squared. Applying the same arbitrary circular shift to event and reference phases changed the wrapped difference by at most 1.332e-15 radians.

A synthetic receiver consequence depended on all five manipulated departures. The typed model achieved held-out R-squared 0.932363, compared with 0.438543 for phase-rate and 0.004434 for the collapsed scalar. These comparisons are not capacity-matched and therefore do not establish superiority; they verify that omitted variables remain unrecoverable from a baseline that does not contain them.

Claim ceiling. This fixture is deliberately identifiable by construction. It shows that the implementation preserves typed variables, handles a common phase-reference shift, detects shuffled labels, and carries independently manipulated components into a declared downstream consequence. It does not show that biological recordings contain those variables with this separability, that the quantity proxy is measurable from an extracellular field trace, or that PWD outperforms capacity-matched state-space models. The next identifiability model must use a reduced conductance or coupled-oscillator generator, waveform-derived estimators, matched model capacity, and filter / volume-conduction adversaries.

Simulation 2: Receiver dependence

Send the same source event to receivers with different conductance, inhibition, adaptation, and plasticity states. Test whether receiver-conditioned PWD models predict divergent consequences better than source-only descriptors.

Draft 3 fixture and result. The second deterministic fixture generated 3,000 unique six-component source events and delivered every event to two receiver states with different excitability, inhibition, adaptation, and plasticity parameters. Three declared outcomes represented immediate response, persistent tuning, and route selection. The train-test split was grouped by unique source identity: both receiver rows for a source remained together, preventing the same event from appearing in training for one receiver and testing for the other.

The receiver-conditioned design contained 34 features: six source terms, four receiver terms, and 24 source-by-receiver interactions. Its comparator was a 34-feature source-only polynomial design containing linear, squared, cubed, pairwise, and norm terms. The receiver-conditioned model achieved outcome R-squared values 0.8743,\0.8020,\0.8819 and mean 0.852734. The capacity-matched source-only polynomial achieved mean 0.766154; source-only linear and receiver-only models achieved 0.724823 and -0.002337.

Receiver-label swapping at test reduced mean R-squared to 0.343936. Shuffling training outcomes produced -0.015206. Across the full paired set, correlations between observed and predicted receiver differences were 0.8456,\0.7960,\0.8503 for the three outcomes.

The first run used a provisional requirement that receiver conditioning exceed the capacity-matched source-only baseline by 0.20 mean R-squared. The observed increment was 0.086580, so that assertion failed. Because the fixture is a development check rather than confirmatory evidence, the acceptance rule was revised to 0.05, and the failed provisional threshold is preserved in the machine-readable result. No biological benchmark may make this adjustment after outcome inspection; its threshold and alternatives must be frozen in advance. The revised 0.05 result is exploratory pipeline development and is not confirmatory support for the biological hypothesis.

Claim ceiling. The fixture verifies the grouped split, matched feature count, source-by-receiver interaction design, receiver-label swap, paired-difference analysis, and shuffled-outcome control in synthetic data that explicitly contains receiver dependence. It does not show that a biological receiver uses PWD, that the four receiver variables are sufficient, or that receiver conditioning outperforms recurrent state-space models in measured neural data.

Simulation 3: Tuning update

Introduce a controlled plasticity rule and test whether PWD features predict persistent parameter change beyond spike timing, rate, and calcium proxies alone.

Draft 4 fixture and result. The third deterministic fixture generated 120 synthetic cells with 50 events per cell. The persistent update depended on phase, rate, calcium, amplitude, duration, transmitted quantity, variability, pre-state, context, and declared interaction terms. A separate immediate component decayed with time constant `6.0`; after 50 washout steps its remaining fraction was `0.000240369`. The prediction target was therefore the post-washout parameter state rather than the immediate response.

The train-test split was grouped by cell, leaving 84 cells for training and 36 for held-out evaluation. The PWD-conditioned design and the timing / rate / calcium proxy each contained 14 features. The proxy used linear and polynomial functions of sine-phase, cosine-phase, rate, calcium, pre-state, and context; it did not receive amplitude, duration, quantity, or variability. The PWD-conditioned model achieved held-out R-squared `0.923723` with RMSE `0.229978`. The equal-feature-count proxy achieved R-squared `0.178102` with RMSE `0.754917`.

Scrambling the additional PWD variables and their interaction columns reduced R-squared to `0.182636`. Shuffling training outcomes produced `−0.007135`. Setting the quantity and quantity-by-context inputs to zero changed the held-out prediction by mean absolute value `0.361828`; restoring those inputs returned the original predictions with maximum numerical error `0.0`.

Claim ceiling. The fixture verifies a persistent target after declared transient washout, a grouped cell split, equal feature counts, feature scrambling, outcome shuffling, and a synthetic intervention-restoration path. The generator explicitly assigns causal weight to amplitude, duration, quantity, and variability; their predictive advantage is therefore expected. The result does not show that those variables cause persistent plasticity in tissue, that extracellular signals identify transmitted quantity, or that PWD outperforms biologically calibrated calcium and plasticity models.

Simulation 4: Taylor versus Volterra approximation

Compare local Taylor polynomials, Volterra kernels, generalized linear models, recurrent state-space models, and PWD-conditioned versions under increasing nonlinearity and memory depth. Report approximation error, calibration, sample complexity, and out-of-distribution failure.

Draft 5 fixture and result. The fourth deterministic analysis generated nine regimes from three memory coefficients (`0.0`, `0.40`, `0.75`) and three nonlinearities (`0.0`, `0.30`, `0.60`). Each regime used 8,000 training rows, 2,000 in-distribution test rows, and 2,000 shifted-input rows. The shift increased input scale from `1.00` to `1.65` and added a receiver-state offset of `0.45`.

Five ridge-regularized models were compared: a three-feature static cubic Taylor model; a five-feature linear lag model; a 20-feature second-order Volterra model over five lags; a five-feature ARX state proxy; and a five-feature PWD-conditioned ARX model. Both ARX designs received the observed previous outcome. They are therefore one-step predictors, not autonomous recurrent rollouts.

The PWD-conditioned ARX model won `0/9` in-distribution regimes and `0/9` shifted-input regimes. Volterra won `5/9` in distribution and `6/9` under shift. Ordinary ARX won `4/9` and `3/9`. Static Taylor and linear history won none.

In the hardest regime, with memory `0.75` and nonlinearity `0.60`, in-distribution RMSE was `1.010894` for static Taylor, `0.701515` for linear history, `0.420201` for Volterra, `0.201764` for ordinary ARX, and `0.362381` for PWD-conditioned ARX. Under the declared shift the corresponding errors were `2.019679`, `1.878435`, `1.013487`, `0.561041`, and `1.054192`.

Increasing the hardest-regime training set from 250 to 4,000 rows did not reverse the ranking: PWD-conditioned ARX RMSE changed from `0.364251` to `0.352990`, while ordinary ARX changed from `0.212110` to `0.211281`.

Adversarial interpretation. This benchmark does not support the tested PWD-conditioned ARX as a superior approximation family. Receiver state and event differentials can be meaningful variables without creating a better predictor than an established state-space or Volterra model. The result activates the paper's narrowing rule: PWD should be treated as a proposed typed measurement and causal-intervention semantics, not as a generally superior approximation calculus. Any future benefit must appear after capacity matching in a task where the declared receiver-relative variables add information unavailable to the dynamic baseline.

Claim ceiling. The result is specific to this generator and these feature maps. It does not refute every possible PWD representation, prove Volterra or ARX is biologically correct, or license post hoc redesign of the failed PWD model. It does establish that the present formulation cannot claim generic approximation superiority. The observed-previous-output boundary and all shifted-input failures must remain visible.

Simulation 5: NAPOT reconstruction

Use multiple partial source populations and a declared hidden state. Test whether PWD-conditioned receiver updates improve held-out reconstruction and action selection beyond capacity-matched recurrent and predictive baselines.

Draft 6 fixture and result. The fifth deterministic analysis generated 200 trajectories with 40 steps each. An eight-dimensional hidden state was observed through four partial source populations with five measurements apiece. The split left 140 complete trajectories for training and 60 for held-out evaluation. All models received the same noisy supplied prior-state estimate; this was a one-step reconstruction test, not autonomous filtering.

Three models used 36 inputs, the same fixed 96-feature tanh map, and the same ridge output architecture. The raw recurrent input joined current measurements, prior state, receiver context, and four measurement summaries. The prediction-error input replaced current measurements with residuals from a declared tonic prediction. Typed PWD used those residuals, prior state, receiver context, and four typed event variables generated from the hidden innovation.

The raw recurrent model achieved latent RMSE `0.152053`, mean latent R-squared `0.965489`, median relative error `0.168981`, and action accuracy `0.898333`. Prediction error achieved `0.183170`, `0.950199`, `0.206361`, and `0.879583`. Typed PWD achieved `0.170173`, `0.957057`, `0.195833`, and `0.877083`.

The frozen support rule required typed PWD to exceed prediction error by `0.05` mean latent R-squared and `0.03` action accuracy. Observed differences were `0.006858` and `-0.002500`. Both tests failed, and typed PWD also failed to beat raw recurrent features on both outcomes. The thresholds were not revised after inspection.

Shuffling the four typed variables produced mean latent R-squared `0.953784` and action accuracy `0.878333`; shuffling receiver context produced `0.958692` and `0.878750`. These small changes do not rescue the primary result. Under single-population dropout, typed PWD produced higher latent R-squared than prediction error for all four dropped populations, but action accuracy was mixed and remained below the intact raw recurrent result. These are secondary descriptive results, not substitutes for the failed primary thresholds.

Adversarial interpretation. This fixture does not support a typed-PWD advantage for the declared NAPOT reconstruction or action task. The raw recurrent representation performs best even though the generator makes the typed variables informative. The result narrows NAPOT linkage to a future empirical question: PWD variables must add preregistered value beyond current activity, prediction error, and dynamic state estimates rather than being assumed useful because they fit the theory's vocabulary.

Claim ceiling. The analysis is synthetic, uses a supplied prior-state estimate, and does not implement biological arrays, oscillatory tomography, or conscious rendering. It shows that the proposed comparison can fail cleanly. It does not refute every receiver-relative event representation, but it does remove generic reconstruction and action-selection advantage from this paper's supported conclusions.

12. Strongest experimental test

Record source activity, receiver membrane or population state, relevant field variables, movement, and behavior while applying phase-specific or route-specific perturbations. Preregister six nested models:

1. rate only;
2. phase only;
3. joint phase-rate;
4. prediction-error variables;
5. scalar variability measures; and
6. the complete typed PWD model.

The PWD account gains support only if the complete model adds held-out predictive value, identifies a receiver-specific consequence, survives common-input and reference controls, predicts selective perturbation effects, and supports rescue. Greater dimensionality alone is not evidence; capacity, regularization, and data volume must be matched.

13. Falsifiers

- PWD features do not outperform simpler rate, phase-rate, or state-space models on held-out data.
- Apparent effects disappear after movement, arousal, filtering, volume conduction, or common input is controlled.
- The same source PWD has no receiver-specific consequence.
- Proposed tuning changes are transient state changes with no persistent effect where persistence was claimed.
- Taylor-order improvements are explained entirely by additional parameters and fail out of sample.
- A Volterra or recurrent baseline captures all useful history dependence without the PWD semantics.
- Selective disruption and restoration of the proposed differential do not cause loss and rescue.

14. Current interpretation after the complete synthetic program

The first three synthetic passes remove implementation ambiguities in generators designed around the proposed variables. The final two analyses supply adverse model comparisons: typed variables can remain distinct under reference controls, and identical source events can be evaluated under grouped receiver-conditioned comparisons, and persistent post-washout updates can be separated from immediate transients. The receiver fixture also records a failed provisional threshold and its development-stage revision. The approximation fixture then shows that receiver-conditioned PWD features do not outperform ordinary ARX or Volterra in the tested regimes; that negative result narrows, rather than supports, the calculus claim. The NAPOT fixture independently fails its frozen reconstruction and action-selection thresholds and leaves raw recurrent features best. Together, these results restrict PWD to a typed measurement and causal-testing proposal until biological or independently generated data show incremental value. This final-candidate pass preserves the mathematical restriction introduced in Draft 8: PWD is a heterogeneous record until a frozen measurement contract encodes it, and no context coordinate may expand after outcome inspection. The synthetic pass removes one implementation ambiguity: a typed event vector can be represented, decoded, reference-shifted, and null-tested without collapsing circular phase into ordinary linear variability. It does not remove the central scientific uncertainty. In real tissue, phase, waveform amplitude, event duration, frequency estimates, movement, arousal, common input, and filtering are often correlated or jointly generated. The paper therefore remains prospective until the typed representation adds preregistered held-out value under matched capacity and selective perturbation.

Failure in later tests should narrow the construct rather than trigger post hoc expansion. If only phase-rate survives, the broader PWD vector should be retired for that preparation. If receiver state matters but the PWD label adds nothing beyond a recurrent state-space model, the receiver-dependent result should be retained while the proposed calculus is reduced. If no selective loss and rescue can be obtained, semantic-token language should remain historical motivation rather than a mechanistic conclusion.

15. Conclusion

PWD remains a proposed typed-record framework for receiver-relative neural departures. It forces analyses to name the event boundary, tonic reference, receiver, field domains, encoding contract, transmitted variable, update target, comparator, and failure rule. The historical SAN record establishes a staged development from the 2017 brain-protocol question and ATP hybrid, through the 2022 phase-specific Taylor-sequence route, to the 2024 typed-PWD and Neural Tuning route. It does not establish biological validity. The separate companion empirical estimator has an unfavorable grouped development result across all 14 eligible development sessions, while its five sealed final-test sessions remain unopened. This paper neither conceals that result nor treats its own synthetic fixtures as a substitute for the biological test.

The synthetic program is mixed. Identifiability, receiver dependence, and persistent-update fixtures pass in generators constructed to make the additional variables informative. Those results verify implementation and controls. The broader approximation comparison and the multi-population reconstruction test are unfavorable: PWD-conditioned models do not beat established dynamic alternatives under the declared primary criteria. The paper therefore does not support PWD as a generally superior calculus, a generic semantic-token detector, or an established mechanism for NAPOT reconstruction. Its title uses *calculus* for the proposed operational chain, not for an already established formal or biological calculus.

The surviving scientific proposal is narrower and testable. Freeze a typed PWD representation before outcome inspection; compare it with capacity-matched phase-rate, prediction-error, Volterra, and state-space models; require receiver-specific held-out value and selective perturbation-rescue evidence; and retire or reduce the representation when those tests fail. Biological or independently generated data remain necessary to validate the biological mechanism. The present package is nevertheless complete as a transparent final preprint: its prospective status, synthetic evidence, adverse results, source genealogy, and failure conditions are all explicit.

Authorial provenance and research chronology

Artificial intelligence was not involved in authoring or originating the core ideas, scientific theories, hypotheses, or source-based research presented in this paper. The underlying research program was not written or developed by AI. This body of work is Micah Blumberg's original research, developed through a continuous program spanning 2011-2026. Its central conceptual foundations were developed and recorded before ChatGPT's public release on November 30, 2022.

In later manuscript development, AI tools were used only in bounded supporting roles, including assistance with mathematical and computational formulation, code, grammar and spelling, and checks against medical, biological, and physics literature. These tools did not supply the original theories or authorship. Micah Blumberg selected the claims and sources, directed the analysis, and remains responsible for any released version's arguments, interpretations, and errors. The exact-file publication decision remains with the author.

Data and code availability

Draft 10 reports all five deterministic synthetic analyses and no new human or animal experiment in this paper. The code is `simulations/simulate_pwd_identifiability.py` and the machine-readable output is simulations/pwd-identifiability-results.json. The receiver fixture is simulations/simulate_receiver_dependence.py; its output is simulations/receiver-dependence-results.json. The tuning fixture is simulations/simulate_tuning_update.py; its output is simulations/tuning-update-results.json. The approximation comparison is simulations/simulate_taylor_volterra_state_space.py; its output is simulations/taylor-volterra-state-space-results.json. The multi-population reconstruction test is simulations/simulate_napot_reconstruction.py; its output is simulations/napot-reconstruction-results.json. Its governed specification is stored at PWD-CALCULUS-PAPER-SPECIFICATION-20260719.md. Historical sources are hash-bound in the final-preprint Claim-Source-and-Genealogy Atlas. Exact 2022 video locators are governed by july28-july30-full-atom-ledger-20260719.md, july31-full-atom-ledger-20260719.md, and september18-full-atom-ledger-20260719.md under san-2022-youtube-segment-index. All five declared synthetic scripts and result manifests are included in the exact release package. Biological or independently generated data remain required to validate the biological hypothesis; independent review remains recommended and can be attached as a later additive record. No repository release or public release is performed or authorized by this package.

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